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Uddeshay Anand

Research Interests

My research interests lie in robot perception, learning and control. I am particularly interested in how robots can learn models of their surroundings to act efficiently and reliably in dynamic environments. My long-term goal is to apply these ideas to intelligent prosthetics and robotic systems that assist people in everyday life.

Education

2024–2026 **M.Tech in Data Analytics**, *Dr. B. R. Ambedkar National Institute of Technology (NIT), Jalandhar, India*, *CGPA: 8.20/10*

Relevant Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Statistical Computing, Databases

2020–2024 **B.Engg. in Mechanical Engineering**, *Panjab University (PU)*, Chandigarh, India, *CGPA: 8.18/10*

Relevant Coursework: Robotics, Mechatronics, Automatic Controls, Vehicle Dynamics, CAD/CAM, Computer Programming

Publications

Journal Articles (Under Review)

MBAN-AFBF: A Multi-Branch Gated Attention Network with Adaptive Spatial–Temporal–Spectral Feature Fusion for Motor Imagery EEG Decoding.U. Anand, K. K. Bharti, M. Sameer.

Contribution: Led the research, conducted the primary experiments and drafted the manuscript under faculty supervision.

Discriminative Transformer Representation Learning from Complementary EEG Descriptors for Emotion Recognition. M. Sameer, K. K. Bharti, **U. Anand**.

Contribution: Assisted with experiments, created manuscript illustrations and drafted the conclusion based on supervisor-provided guidance.

Mobile Node Localization in Wireless Sensor Networks Using Graph Neural Networks and Deep Learning Models. A. Sharma, A. S. Shamkuwar, P. Bhardwaj, R. Kumar, N. Kumar,**U. Anand**, M. Singh.

Contribution: Conducted experiments and ablation studies to evaluate node localization models.

Conference Papers (Accepted)

Multiband Attention Network for Motor Imagery EEG Classification.U. Anand, K. K. Bharti, M. Sameer. *MBCC 2026, IIT Mandi.* [Paper] [Code]

Contribution: Led the research, conducted the primary experiments and drafted the manuscript under faculty supervision.

ECOS-EEG: Evolutionary Channel Optimization for Subject-Independent EEG Emotion Recognition. M. Sameer, K. K. Bharti, **U. Anand**. *ICCEEE 2026, NIT Uttarakhand.* [Paper]
Conducted experiments for valence classification and contributed to the preparation of Fig. 1

Spatio-Temporal Node Localization in Wireless Sensor Networks Using Hybrid Recurrent-Graph Neural Network. A. Sharma, A. S. Shamkuwar, **U. Anand**, R. Kumar, N. Kumar, M. Singh. *ICSCCC 2026, NIT Jalandhar.* [Paper]
Contribution: Conducted simulation experiments and evaluated four graph-based and recurrent models across 15 mobility and noise settings.

Academic Appointments

Apr 2026 **Reviewer, ICICST 2026**, *Department of Information Technology, NIT Jalandhar*
Served as a peer reviewer for the International Conference (April 3–5, 2026)

Feb 2025 – May 2025 **Teaching Assistant – Machine Learning**, *NIT Jalandhar*
Assisted in course delivery, assignment design and evaluation for a graduate-level ML course

Aug 2024 – Nov 2024 **Teaching Assistant – Python Programming**, *NIT Jalandhar*
Assisted in course delivery and lab sessions for a graduate-level Python Programming course

Research & Industry Experience

Mar 2026 – Present **Data Science Intern**, *Oceaneering*
Promoted Sep 2026; Data Science Intern, Mar–Aug 2026
- Developed a visual similarity retrieval pipeline for video frames using DINOv2 embeddings and FAISS vector indexing.
- Deployed DINOv2 inference using NVIDIA Triton Inference Server to support the Image Query App.
- Configured dynamic batching and evaluated inference performance, achieving a 33% increase in throughput.

Jan 2024 – May 2024 **Machine Learning Intern**, *Artificial Limbs Manufacturing Corporation of India (ALIMCO)*
- Contributed to an ML-assisted prosthetic-control concept integrating EMG-based user intent with flex-sensor position feedback.
- Developed a regression pipeline to predict prosthetic-ankle position feedback from EMG signals and current flex-sensor readings; processed biomedical recordings using Python, NumPy, pandas and SciPy
- Trained and evaluated Linear Regression and Random Forest models, achieving $R^2 \approx 0.98$ and cross-validation scores above 0.96.

May 2021 – Aug 2021 **CAD Automation Intern**, *Larsen & Toubro*
- Developed an offline robot program generation workflow for a KUKA robot, automating trajectory creation from CAD/STL geometry using coordinate transformations and robot kinematics
- Built a Python-based pipeline to extract geometric information from SolidWorks models and export waypoints/robot commands for downstream robotic execution

Relevant Projects

Remaining Useful Life (RUL) Prediction for Aircraft Engines
- Modeled degradation behavior on the NASA C-MAPSS turbofan dataset; Random Forest achieved best performance (RMSE: 29.56 cycles) after feature reduction and EDA

3-Axis CNC Milling Machine – DST Sponsored Project

- Designed and developed a lab-scale 3-axis CNC milling machine (team of 5), funded by DST, Chandigarh (INR 50,000), using NEMA23 stepper motors and Mach3-based motion control

Portable Wrapping Machine – Innovative Product Design

- Designed and fabricated a portable wrapping machine with integrated power backup, handling component selection, power calculations and system testing

Technical Skills

Languages	Python, C++, MATLAB, SQL
ML/DL	PyTorch, TensorFlow, scikit-learn, Keras, OpenCV
Hardware	ROS/ROS2, Arduino, SolidWorks, CAD/CAM
Scientific	NumPy, pandas, SciPy, Matplotlib
Systems	Git, Linux, Docker, Triton Inference Server

Achievements & Honors

GATE 2024 Qualified – 91.73 percentile, Computer Science & Engineering

AICTE GATE Scholarship – Merit-based stipend for M.Tech, awarded by All India Council for Technical Education (AICTE), Ministry of Education, Government of India

Full Tuition Fee Waiver – Awarded for ranking in the top 6 of the cohort, covering 4 semesters (1st and final year) of undergraduate

Amazon ML Challenge 2025 – Top 200 of 7,000+ participants nationwide

References

Available on request